

```

import math
import matplotlib.pyplot as plt
import numpy as np

# =====
# DADOS
# =====

x = [0, 2, 1, 3]
y = [0, 0, 1, 1]

bars = [
    (0, 1),
    (0, 2),
    (1, 2),
    (1, 3),
    (2, 3)
]

Fx = [0, 0, 0, 0]
Fy = [0, 0, -1000, 0]

N = len(x)
B = len(bars)

unknowns = B + 3
eqs = 2 * N

A = [[0.0 for _ in range(unknowns)] for _ in range(eqs)]
b = [0.0 for _ in range(eqs)]

# =====
# BARRAS
# =====

for k, (i, j) in enumerate(bars):

    dx = x[j] - x[i]
    dy = y[j] - y[i]

    L = math.sqrt(dx**2 + dy**2)

    cx = dx / L
    cy = dy / L

    A[2*i][k] += cx
    A[2*i+1][k] += cy

```

```

A[2*i][k] -= cx
A[2*i+1][k] -= cy

# =====
# APOIOS
# =====

A[0][B + 0] = 1
A[1][B + 1] = 1
A[3][B + 2] = 1

# =====
# CARGAS
# =====

for i in range(N):
    b[2*i] = -Fx[i]
    b[2*i+1] = -Fy[i]

# =====
# GAUSS
# =====

def gauss(A, b):

    n = len(b)
    M = [A[i] + [b[i]] for i in range(n)]

    for k in range(n):

        max_row = max(range(k, n), key=lambda i: abs(M[i][k]))
        M[k], M[max_row] = M[max_row], M[k]

        pivot = M[k][k]
        if abs(pivot) < 1e-12:
            raise ValueError("Sistema singular")

        for j in range(k, n + 1):
            M[k][j] /= pivot

        for i in range(k + 1, n):
            f = M[i][k]
            for j in range(k, n + 1):
                M[i][j] -= f * M[k][j]

    xsol = [0.0] * n

    for i in range(n - 1, -1, -1):

```

```

        xsol[i] = M[i][n]
        for j in range(i + 1, n):
            xsol[i] -= M[i][j] * xsol[j]

    return xsol

sol = gauss(A, b)

forces = sol[:B]

# =====
# ESTILO VISUAL DOS GRÁFICOS
# =====

forces_abs = np.array([abs(f) for f in forces])
max_f = max(forces_abs)

# =====
# GRÁFICO 1 — TRELIÇA CIRCULAR (NOVO ESTILO)
# =====

plt.figure(figsize=(6, 6))

for i, j in bars:
    plt.plot([x[i], x[j]], [y[i], y[j]], 'k-', alpha=0.5)

plt.scatter(x, y, c='black', s=80)

for i in range(N):
    plt.text(x[i], y[i]+0.05, f'{i}')

plt.title("Trelença (Estilo Rede)")
plt.axis('equal')
plt.grid(True)

# =====
# GRÁFICO 2 — CÍRCULOS PROPORCIONAIS (FORÇA)
# =====

plt.figure(figsize=(6, 6))

for k, (i, j) in enumerate(bars):

    f = forces[k]

    size = 800 * (abs(f) / max_f)

```

```

if f >= 0:
    color = "blue"
else:
    color = "red"

mx = (x[i] + x[j]) / 2
my = (y[i] + y[j]) / 2

plt.scatter(mx, my, s=size, c=color, alpha=0.6)

plt.plot([x[i], x[j]], [y[i], y[j]], 'k--', alpha=0.3)

plt.title("Intensidade de Força (Bubbles)")
plt.axis('equal')
plt.grid(True)

# =====
# GRÁFICO 3 — HEATMAP DE BARRAS (VERTICAL)
# =====

plt.figure(figsize=(6, 4))

bars_id = np.arange(B)

colors = ['blue' if f > 0 else 'red' for f in forces]

plt.barh(bars_id, forces, color=colors)

plt.title("Forças nas Barras (Bar Chart Horizontal)")
plt.xlabel("Força (N)")
plt.ylabel("Barras")
plt.grid(True)

# =====
# GRÁFICO 4 — DISTRIBUIÇÃO ESTATÍSTICA
# =====

plt.figure(figsize=(6, 4))

plt.hist(forces, bins=6, color='gray', edgecolor='black')

plt.title("Distribuição das Forças")
plt.xlabel("Força (N)")
plt.ylabel("Frequência")
plt.grid(True)

# =====
# MOSTRAR

```

```
# =====
```

```
plt.tight_layout()  
plt.show()
```